Precept 7 Outline - KEY

ECO 202 TAs

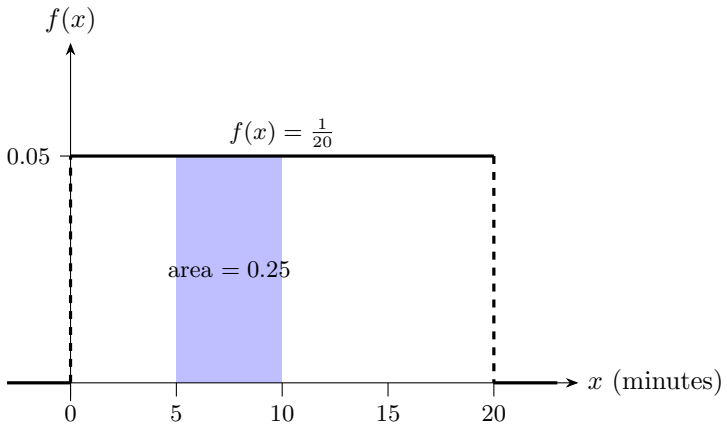
November 5, 2026

1 Weekly Check In

1. Any questions about the class, comments?
2. Exam 3 coming up this Monday

2 Precept practice problems

1. You walk up to a bus stop knowing a bus will arrive within the next 20 minutes, and every moment in that window is equally likely. Let X be your waiting time in minutes, so $X \sim \text{Uniform}(0, 20)$.



(a) Write down the density $f(x)$ for all real x. Then check that the total area under f equals 1.

Solution. The density is constant on the window and zero elsewhere:

$$f(x) = \begin{cases} \frac{1}{20}, & 0 \le x \le 20, \\ 0, & \text{otherwise.} \end{cases}$$

The region under f is a rectangle of width 20 and height $\frac{1}{20}$, so the total area is $20 \cdot \frac{1}{20} = 1$. The height is *chosen* to be $1/(b-a)$ precisely so that this area equals 1.

(b) The density at $x = 5$ is $f(5) = 0.05$. Is this the probability that the bus arrives at exactly 5 minutes? What does the height $f(x)$ actually measure (think about its units)?

Solution. No. The height $f(5) = 0.05$ is *not* $\mathbb{P}(X = 5)$. Its units are probability *per minute*: it measures how concentrated the probability is near $x = 5$. Over a short interval of length dx near 5, the probability is approximately $f(5)\,dx$, so a one-second window ($dx = \frac{1}{60}$ min) has probability about $0.05/60 \approx 0.0008$. Probability comes from height *times* width, i.e. from area.

(c) Find $\mathbb{P}(5 \le X \le 10)$ by computing the shaded area in the figure.

Solution. The shaded region is a rectangle of width $10 - 5 = 5$ and height 0.05:

$$\mathbb{P}(5 \le X \le 10) = \int_5^{10} \frac{1}{20}\, dx = 5 \cdot \frac{1}{20} = \frac{1}{4} = 0.25.$$

(d) For $0 < h \le 15$, compute $\mathbb{P}(5 \le X \le 5 + h)$. Let $h \to 0$ to conclude what $\mathbb{P}(X = 5)$ is. Then explain why $\mathbb{P}(X < 5) = \mathbb{P}(X \le 5)$. Does "probability 0" mean "impossible"?

Solution. For $h > 0$ (small enough that $5 + h \le 20$),

$$\mathbb{P}(5 \le X \le 5 + h) = h \cdot \frac{1}{20} = \frac{h}{20}.$$

The event $\{X = 5\}$ is contained in $\{5 \le X \le 5 + h\}$ for every $h > 0$, so $\mathbb{P}(X = 5) \le h/20$ for every $h > 0$. Letting $h \to 0$ gives $\mathbb{P}(X = 5) = 0$. Geometrically, a single point is a rectangle of width 0, which has area 0 no matter how tall it is.

The same holds at every point, so $\mathbb{P}(X \le 5) = \mathbb{P}(X < 5) + \mathbb{P}(X = 5) = \mathbb{P}(X < 5)$; endpoints of an interval do not matter for a continuous random variable.

Probability 0 does *not* mean impossible: the bus will arrive at *some* exact time, and that time had probability 0 of being the one that occurred. With uncountably many possible values, probability 0 for each point is consistent with total probability 1 over the interval.

(e) Let $Y \sim \text{Uniform}(0, \frac{1}{2})$. Find its density height and $\mathbb{P}(0.1 \le Y \le 0.2)$. How can a density be larger than 1 when probabilities cannot?

Solution. The width of the support is $\frac{1}{2}$, so the height must be $1/(1/2) = 2$, i.e. $f_Y(y) = 2$ on $[0, \frac{1}{2}]$. Then

$$\mathbb{P}(0.1 \le Y \le 0.2) = (0.2 - 0.1) \cdot 2 = 0.2.$$

A density of 2 is fine because it is a probability *per unit of* y, not a probability. Only areas are probabilities, and the total area is $\frac{1}{2} \cdot 2 = 1$. Height can exceed 1 whenever the probability is packed into a region narrower than 1 unit.

2. **(Practice 10, Question 5)**

Two weekly cost shocks A and B, measured in hundreds of dollars, satisfy

$$\text{Var}(A) = 16, \qquad \text{Var}(B) = 9, \qquad \text{Cov}(A, B) = -6.$$

(a) Calculate $\text{SD}(A)$, $\text{SD}(B)$, and $\text{Corr}(A, B)$.

Solution. The standard deviations are $\text{SD}(A) = \sqrt{16} = 4$ and $\text{SD}(B) = \sqrt{9} = 3$ (hundreds of dollars), so

$$\text{Corr}(A, B) = \frac{\text{Cov}(A, B)}{\text{SD}(A)\,\text{SD}(B)} = \frac{-6}{(4)(3)} = -0.50.$$

Correlation is unitless.

(b) Calculate $\text{Var}(A + B)$. Compare it with the answer obtained by incorrectly omitting covariance.

Solution. Keeping the covariance term,

$$\text{Var}(A + B) = \text{Var}(A) + \text{Var}(B) + 2\,\text{Cov}(A, B) = 16 + 9 + 2(-6) = 13.$$

Omitting covariance would incorrectly give $16 + 9 = 25$. Because the shocks are negatively related, they partly offset each other and the sum is less variable than the naive calculation suggests.

(c) If A and B were independent with the same marginal variances, what would $\text{Cov}(A, B)$ and $\text{Var}(A + B)$ be? Which implication are you using?

Solution. Independence implies zero covariance, so $\text{Cov}(A, B) = 0$ and

$$\text{Var}(A + B) = 16 + 9 + 2(0) = 25.$$

The implication used is *independence* $\Rightarrow$ *zero covariance*. Zero covariance alone would give the same variance formula, but it would not establish independence.

Now let U take values $-2, -1, 1, 2$ with probability $1/4$ each and let $V = \mathbf{1}\{|U| = 2\}$.

(d) Calculate $\mathbb{E}[U]$, $\mathbb{E}[V]$, $\mathbb{E}[UV]$, and $\mathrm{Cov}(U,V)$.

Solution. The four equally likely outcomes are

u	-2	-1	1	2
$v = \mathbf{1}\{\lvert u\rvert = 2\}$	1	0	0	1

Then $\mathbb{E}[U] = \frac{1}{4}(-2-1+1+2) = 0$ and $\mathbb{E}[V] = \frac{1}{4}(1+0+0+1) = \frac{1}{2}$, and

$$\mathbb{E}[UV] = \tfrac{1}{4}\big((-2)(1) + (-1)(0) + (1)(0) + (2)(1)\big) = 0.$$

Therefore

$$\mathrm{Cov}(U,V) = \mathbb{E}[UV] - \mathbb{E}[U]\,\mathbb{E}[V] = 0 - (0)\left(\tfrac{1}{2}\right) = 0.$$

(e) Show directly that U and V are dependent despite their zero covariance.

Solution. V is completely determined by U. For example,

$$\mathbb{P}(V = 1 \mid U = 2) = 1 \;\neq\; \tfrac{1}{2} = \mathbb{P}(V = 1),$$

so knowing U changes the distribution of V. The variables are dependent even though $\mathrm{Cov}(U,V) = 0$. This counterexample shows that zero covariance does *not* imply independence.

(f) Calculate $\mathrm{Var}(U+V)$ with the variance-of-a-sum formula and verify it from the distribution of $U+V$.

Solution. The variances are

$$\mathrm{Var}(U) = \mathbb{E}[U^2] = \frac{4+1+1+4}{4} = 2.5, \qquad \mathrm{Var}(V) = \tfrac{1}{2}\left(1 - \tfrac{1}{2}\right) = 0.25.$$

Because the covariance is zero,

$$\mathrm{Var}(U+V) = 2.5 + 0.25 + 2(0) = 2.75.$$

Direct check. Across the four outcomes, $U+V$ equals -1, -1, 1, 3. Its mean is 0.5 and its second moment is $\frac{1+1+1+9}{4} = 3$, so

$$\mathrm{Var}(U+V) = 3 - (0.5)^2 = 2.75,$$

which agrees with the formula.

3. **(Practice 13, Question 4)**

A finite population has serial numbers $1, 2, 3, 4, 5$, so the unknown-target value in this fully visible teaching case is $m = 5$. Select an SRS of $n = 2$ distinct serial numbers without replacement. There are $\binom{5}{2} = 10$ equally likely samples.

Consider the mean-based rule $T_{\text{mean}} = 2\bar{X} - 1$, the raw maximum M, and the corrected-maximum rule $T_{\max} = \frac{3}{2}M - 1$.

(a) List all ten possible samples.

Solution. The ten samples are $\{1,2\}$, $\{1,3\}$, $\{1,4\}$, $\{1,5\}$, $\{2,3\}$, $\{2,4\}$, $\{2,5\}$, $\{3,4\}$, $\{3,5\}$, $\{4,5\}$, each with probability $1/10$.

(b) For each sample, calculate T_{mean}, M, and $T_{\max}$.

Solution.

Sample	T_{mean}	M	$T_{\max}$
$\{1,2\}$	2	2	2
$\{1,3\}$	3	3	3.5
$\{1,4\}$	4	4	5
$\{1,5\}$	5	5	6.5
$\{2,3\}$	4	3	3.5
$\{2,4\}$	5	4	5
$\{2,5\}$	6	5	6.5
$\{3,4\}$	6	4	5
$\{3,5\}$	7	5	6.5
$\{4,5\}$	8	5	6.5

For example, for $\{2,5\}$ we have $\bar{X} = 3.5$, so $T_{\text{mean}} = 2(3.5) - 1 = 6$; $M = 5$; and $T_{\max} = \frac{3}{2}(5) - 1 = 6.5$.

(c) Construct the sampling distribution of each rule.

Solution. Tally the columns of the table in (b); each sample has probability $1/10$, so the probability is the count divided by 10.

T_{mean} value	2	3	4	5	6	7	8
count	1	1	2	2	2	1	1
M value	2	3	4	5			
count	1	2	3	4			
$T_{\max}$ value	2	3.5	5	6.5			
count	1	2	3	4			

Each row of counts sums to 10.

(d) Calculate the expectation and bias of all three rules for $m = 5$.

Solution.

$$\mathbb{E}[T_{\text{mean}}] = \frac{2+3+2(4)+2(5)+2(6)+7+8}{10} = 5, \qquad \text{Bias}(T_{\text{mean}}) = 5-5 = 0,$$

$$\mathbb{E}[M] = \frac{1(2)+2(3)+3(4)+4(5)}{10} = 4, \qquad \text{Bias}(M) = 4-5 = -1,$$

$$\mathbb{E}[T_{\max}] = \frac{1(2)+2(3.5)+3(5)+4(6.5)}{10} = 5, \qquad \text{Bias}(T_{\max}) = 5-5 = 0.$$

The mean-based rule and the corrected maximum are unbiased for $m = 5$ in this model; the raw maximum underestimates m on average.

(e) Calculate the variance and MSE of the two unbiased rules.

Solution. Because both rules are unbiased, MSE = Var. For the mean-based rule, the squared deviations from 5 sum to $9+4+2(1)+2(0)+2(1)+4+9 = 30$, so

$$\text{Var}(T_{\text{mean}}) = \text{MSE}(T_{\text{mean}}) = \frac{30}{10} = 3.$$

For the corrected maximum,

$$1(2-5)^2 + 2(3.5-5)^2 + 3(5-5)^2 + 4(6.5-5)^2 = 22.5,$$

so

$$\text{Var}(T_{\max}) = \text{MSE}(T_{\max}) = \frac{22.5}{10} = 2.25.$$

(f) Which unbiased rule has smaller MSE in this model? Explain why this finite example does not establish the same ranking under every alternative sampling mechanism or loss.

Solution. The corrected maximum has the smaller MSE ($2.25 < 3$). The comparison is conditional on this design (SRS without replacement, $n = 2$), this target ($m = 5$), and squared-error loss. A different sampling mechanism changes the possible samples and their probabilities, and a different loss function can rank the rules differently, so one finite enumeration does not establish the same ordering in general.

(g) Explain why changing to independent sampling with replacement requires a new calculation rather than the same corrected-maximum formula.

Solution. With replacement, the possible samples include repeated serial numbers, and their probabilities differ from the ten equally likely without-replacement samples. In particular, the distribution of the maximum M changes, so $\mathbb{E}[M]$ changes. The correction $\frac{3}{2}M - 1$ was chosen so that the estimator has expectation m under the without-replacement design, so the expectation must be derived again for the new design before any bias correction is valid.

4. A bag holds three tickets labeled 2, 4, and 6. A ticket is chosen at random, with each ticket equally likely. Let X be the number on one randomly chosen ticket, with population mean $\mu = \mathbb{E}[X]$ and population variance $\sigma^2 = \text{Var}(X)$.

We will study two statistics computed from a sample of two tickets, X_1 and X_2:

$$\overline{X} = \frac{X_1 + X_2}{2} \qquad \text{and} \qquad M = \max(X_1, X_2)$$

The goal is to see when the usual formula $\text{Var}(\overline{X}) = \sigma^2/n$ gives the right answer, and when it does not.

(a) Calculate μ and σ^2 for a single draw.

Solution. The three values are equally likely, so

$$\mu = \frac{2+4+6}{3} = 4$$

and

$$\sigma^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{8}{3}$$

(b) Draw a ticket, record it, put it back, and draw again, so that X_1 and X_2 are independent. List all nine equally likely ordered samples (X_1, X_2) and, for each, calculate $\overline{X}$ and M.

Solution. There are $3 \times 3 = 9$ ordered samples, each with probability $1/9$:

Ordered sample	$(2,2)$	$(2,4)$	$(2,6)$	$(4,2)$	$(4,4)$	$(4,6)$	$(6,2)$	$(6,4)$	$(6,6)$
$\overline{X}$	2	3	4	3	4	5	4	5	6
M	2	4	6	4	4	6	6	6	6

(c) Construct the sampling distribution of $\overline{X}$ and of M, and verify that each set of probabilities sums to one.

Solution. Count how many of the nine samples give each value, and divide by 9:

$\overline{x}$	2	3	4	5	6
$\mathbb{P}(\overline{X} = \overline{x})$	$1/9$	$2/9$	$3/9$	$2/9$	$1/9$

m	2	4	6
$\mathbb{P}(M = m)$	$1/9$	$3/9$	$5/9$

Both sets of probabilities are nonnegative and sum to one: $1+2+3+2+1 = 9$ and $1+3+5 = 9$.

(d) Using the sampling distribution of $\overline{X}$, calculate $\mathbb{E}[\overline{X}]$ and $\mathrm{Var}(\overline{X})$.

Solution. The expectation is

$$\mathbb{E}[\overline{X}] = \frac{2(1)+3(2)+4(3)+5(2)+6(1)}{9} = \frac{36}{9} = 4$$

which equals μ. For the variance,

$$\mathrm{Var}(\overline{X}) = \frac{1(2-4)^2+2(3-4)^2+3(4-4)^2+2(5-4)^2+1(6-4)^2}{9} = \frac{4+2+0+2+4}{9} = \frac{4}{3}$$

(e) Using the sampling distribution of M, calculate $\mathbb{E}[M]$ and $\mathrm{Var}(M)$.

Solution. The expectation is

$$\mathbb{E}[M] = \frac{2(1)+4(3)+6(5)}{9} = \frac{44}{9} \approx 4.89$$

For the variance, first

$$\mathbb{E}[M^2] = \frac{4(1)+16(3)+36(5)}{9} = \frac{232}{9}$$

so

$$\mathrm{Var}(M) = \frac{232}{9} - \left(\frac{44}{9}\right)^2 = \frac{2088-1936}{81} = \frac{152}{81} \approx 1.88$$

(f) Calculate σ^2/n with $n = 2$. Compare it with $\mathrm{Var}(\overline{X})$ and with $\mathrm{Var}(M)$. For which statistic does the formula give the right answer?

Solution. We have

$$\frac{\sigma^2}{n} = \frac{8/3}{2} = \frac{4}{3}$$

This equals $\mathrm{Var}(\overline{X}) = 4/3$ exactly. It does *not* equal $\mathrm{Var}(M) = 152/81 \approx 1.88$. The shortcut σ^2/n is derived specifically for the sample mean of independent draws, so it cannot be applied automatically to another statistic. For M, the variance has to be computed from its own sampling distribution.

(g) Now draw only *one* ticket X and record its value twice, giving the "sample" (X, X). Let $\overline{X} = (X+X)/2$. Find $\mathrm{Var}(\overline{X})$. Does it equal $\sigma^2/2$? Explain.

Solution. Here $\overline{X} = (X+X)/2 = X$, which equals 2, 4, or 6 with probability $1/3$ each. Hence

$$\mathrm{Var}(\overline{X}) = \mathrm{Var}(X) = \frac{8}{3}$$

The formula $\sigma^2/2 = 4/3$ is only half of the true value. The reason is that the two recorded values are not independent: they are identical, so their correlation is 1 and

$$\mathrm{Var}\left(\frac{X+X}{2}\right) = \frac{1}{4}\left[\mathrm{Var}(X) + \mathrm{Var}(X) + 2\,\mathrm{Cov}(X,X)\right] = \frac{1}{4}\left[4\sigma^2\right] = \sigma^2$$

The covariance term is as large as possible, and it cancels the benefit of averaging. Recording the same draw twice adds no new information.

(h) Draw two tickets *without* replacement. List the possible samples (ignoring order), find the sampling distribution of $\overline{X}$, and calculate $\mathbb{E}[\overline{X}]$ and $\mathrm{Var}(\overline{X})$. Compare with $\sigma^2/2$.

Solution. There are three equally likely unordered samples:

Sample	$\{2,4\}$	$\{2,6\}$	$\{4,6\}$
$\overline{X}$	3	4	5
Probability	1/3	1/3	1/3

Then $\mathbb{E}[\overline{X}] = (3+4+5)/3 = 4 = \mu$. The variance is

$$\mathrm{Var}(\overline{X}) = \frac{(3-4)^2 + (4-4)^2 + (5-4)^2}{3} = \frac{2}{3}$$

This is smaller than $\sigma^2/2 = 4/3$. Without replacement, the second draw depends on the first (once a ticket is removed it cannot appear again), so the draws are negatively dependent and the variance of the mean is reduced. The correct formula includes a finite-population correction, with population size $N = 3$ and sample size $n = 2$:

$$\frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{4}{3} \cdot \frac{1}{2} = \frac{2}{3}$$

which matches.

(i) Fill in the table: for each scenario, does $\mathrm{Var} = \sigma^2/n = 4/3$ hold? Then state the two conditions that must both be satisfied before using σ^2/n.

Scenario	Statistic	Variance	Equals 4/3?
2 draws, with replacement	$\overline{X}$		
2 draws, with replacement	M		
1 draw, copied twice	$\overline{X}$		
2 draws, without replacement	$\overline{X}$		

Solution.

Scenario	Statistic	Variance	Equals 4/3?
2 draws, with replacement	$\overline{X}$	4/3	Yes
2 draws, with replacement	M	$152/81 \approx 1.88$	No
1 draw, copied twice	$\overline{X}$	8/3	No (too large)
2 draws, without replacement	$\overline{X}$	2/3	No (too small)

The formula $\mathrm{Var} = \sigma^2/n$ requires both that (i) the statistic is the *sample mean*, and (ii) the observations are *independent* (equivalently, all covariance terms in the variance of the sum are zero). When either condition fails, return to the definition: list the possible samples, compute the statistic for each, attach probabilities, and compute the variance of that sampling distribution.

3 Extra practice problems

1. Let Y take the values 0, 2, 4 with probabilities 0.30, 0.45, 0.25, and let $Y_1, \ldots, Y_{50}$ be independent random variables with the same distribution as Y.

   (a) Calculate $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$.

   Solution. $\mathbb{E}[Y] = 0(0.30) + 2(0.45) + 4(0.25) = 1.9$ and $\mathbb{E}[Y^2] = 4(0.45) + 16(0.25) = 5.8$, so

   $$\mathrm{Var}(Y) = 5.8 - (1.9)^2 = \boxed{2.19}.$$

   (b) Use the central limit theorem, without a continuity correction, to approximate $\mathbb{P}\big(\sum_{i=1}^{50} Y_i < 100\big)$. You may use $\Phi(0.48) \approx 0.6844$. Say which steps are exact and which are approximate.

   Solution. Exactly (linearity, and independence for the variance), $\mathbb{E}\big[\sum Y_i\big] = 50(1.9) = 95$ and $\mathrm{Var}\big(\sum Y_i\big) = 50(2.19) = 109.5$, so the standard deviation is ≈ 10.46. The CLT supplies the Normal shape (the approximate step):

   $$\mathbb{P}\Big(\sum Y_i < 100\Big) \approx \mathbb{P}\Big(Z < \tfrac{100-95}{10.46}\Big) = \Phi(0.48) \approx \boxed{0.684}.$$

   Equivalently, $\overline{Y}_{50} < 2$ with $\overline{Y}_{50}$ approximately $\mathsf{N}(1.9,\ 2.19/50)$.

2. Let Y take the values 1, 2, 3 with probabilities 0.60, 0.10, 0.30, and let $Y_1, \ldots, Y_{100}$ be independent random variables with the same distribution as Y.

(a) Calculate $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$.

Solution. $\mathbb{E}[Y] = 0.6 + 0.2 + 0.9 = 1.7$ and $\mathbb{E}[Y^2] = 0.6 + 0.4 + 2.7 = 3.7$, so $\mathrm{Var}(Y) = 3.7 - (1.7)^2 = \boxed{0.81}$.

(b) Give the approximate distribution of $\sum_{i=1}^{100} Y_i$ and say what justifies it.

Solution. The mean $100(1.7) = 170$ and variance $100(0.81) = 81$ are exact under independence. The Normal shape is the CLT approximation: $\sum Y_i \dot{\sim} \mathsf{N}(170, 81)$, with standard deviation 9.

(c) Find the approximate interquartile range of $\sum_{i=1}^{100} Y_i$. Use that the 75th percentile of the standard Normal distribution is about 0.67.

Solution. The quartiles are $170 \pm 0.67(9) = 170 \pm 6.03$, that is, about 163.97 and 176.03, so

$$\mathrm{IQR} \approx \boxed{12.06}.$$

The IQR of a Normal distribution depends only on its spread, not its center.

3. A large population has unknown mean μ and known standard deviation $\sigma = 7$. Take two independent simple random samples: one of size $n_1 = 100$ with sample mean $\overline{X}_1$, and one of size $n_2 = 400$ with sample mean $\overline{X}_2$.

(a) Let $Y = \frac{1}{2}\overline{X}_1 + \frac{1}{2}\overline{X}_2$. Find $\mathbb{E}[Y]$ and $\mathrm{SD}(Y)$, then approximate $\mathbb{P}(Y - \mu > 0.2)$. You may use $\Phi(0.51) \approx 0.6950$.

Solution. By linearity $\mathbb{E}[Y] = \mu$. By independence of the two samples,

$$\mathrm{Var}(Y) = \tfrac{1}{4} \cdot \frac{49}{100} + \tfrac{1}{4} \cdot \frac{49}{400} = 0.153125, \qquad \mathrm{SD}(Y) \approx 0.391.$$

Each sample mean is approximately Normal (CLT), so Y is approximately Normal, and

$$\mathbb{P}(Y - \mu > 0.2) \approx \mathbb{P}\Big(Z > \tfrac{0.2}{0.391}\Big) = 1 - \Phi(0.51) \approx \boxed{0.305}.$$

(b) For a number w, let $T_w = w\overline{X}_1 + (1-w)\overline{X}_2$. Show that T_w is unbiased for μ for every w.

Solution. By linearity of expectation, $\mathbb{E}[T_w] = w\mathbb{E}[\overline{X}_1] + (1-w)\mathbb{E}[\overline{X}_2] = w\mu + (1-w)\mu = \mu$. Only linearity is used, so the result holds for every w.

(c) Compare $\mathrm{Var}(T_{1/2})$ and $\mathrm{Var}(T_{1/5})$. Which estimator is preferable, and why?

Solution. By independence, $\mathrm{Var}(T_w) = w^2 \dfrac{49}{100} + (1-w)^2 \dfrac{49}{400}$. Then

$$\mathrm{Var}(T_{1/2}) = 0.153125, \qquad \mathrm{Var}(T_{1/5}) = 0.04(0.49) + 0.64(0.1225) = 0.098.$$

Both are unbiased, so MSE = Var and $T_{1/5}$ is preferable. The weight $w = n_1/(n_1+n_2) = 1/5$ gives the average of all 500 observations, with every observation weighted equally; the simple average of the two sample means over-weights the smaller sample.

4. On a hot day there are 8,000 cicadas in audible range. At any given moment a cicada either sings (probability 0.16) or does not (probability 0.84), and cicadas decide independently. The overall noise level is proportional to the number of singing cicadas.

(a) Without computing anything, give a statistical reason to expect the fraction of singing cicadas to be about 0.16.

Solution. The fraction is a sample mean of 8,000 independent Bernoulli(0.16) indicators. By the law of large numbers, a sample mean concentrates around its expectation, here 0.16, when n is large.

(b) You listen at 9:00am. What is the approximate probability that the fraction of singing cicadas exceeds 0.17? You may use $\Phi(2.44) \approx 0.9927$.

Solution. Let $\widehat{P}$ be the fraction. Exactly, $\mathbb{E}[\widehat{P}] = 0.16$ and $\mathrm{SE}(\widehat{P}) = \sqrt{0.16(0.84)/8000} \approx 0.0041$. By the CLT,

$$\mathbb{P}(\widehat{P} > 0.17) \approx \mathbb{P}\Big(Z > \tfrac{0.01}{0.0041}\Big) = 1 - \Phi(2.44) \approx \boxed{0.007}.$$

With $n = 8000$ the half-unit continuity adjustment changes $\widehat{P}$ by only 0.00006, which is negligible.

(c) You listen again at 9:10am, and behavior at 9:10 is independent of behavior at 9:00. What is the approximate probability that the noise at 9:10 is more than 2% larger than at 9:00? (Hint: for positive numbers A and B, $B/A > 1.02$ whenever $B - 1.02A > 0$. You may use $\Phi(0.55) \approx 0.7088$.)

Solution. Let $\widehat{P}_1, \widehat{P}_2$ be the two fractions (noise is proportional to them). We need $\mathbb{P}(\widehat{P}_2 - 1.02\,\widehat{P}_1 > 0)$. The difference is approximately Normal (CLT, independence), with

$$\mathbb{E} = 0.16 - 1.02(0.16) = -0.0032, \qquad \text{Var} = \frac{0.16(0.84)}{8000}\left(1 + 1.02^2\right) \Rightarrow \text{SD} \approx 0.00585.$$

Therefore

$$\mathbb{P}(\widehat{P}_2 - 1.02\,\widehat{P}_1 > 0) \approx \mathbb{P}\Big(Z > \tfrac{0.0032}{0.00585}\Big) = 1 - \Phi(0.55) \approx \boxed{0.29}.$$

5. Let $X \sim \text{Binomial}(n, p)$ and define $Y = n - X$.

(a) Is Y also binomially distributed? Explain.

Solution. Yes. If X counts successes in n independent trials, then $n - X$ counts failures, so $Y \sim \text{Binomial}(n,\, 1-p)$.

(b) Calculate $\text{Cov}(X, Y)$. (The answer depends on n and p.) Explain the sign.

Solution. Since $Y - \mathbb{E}[Y] = -(X - \mathbb{E}[X])$,

$$\text{Cov}(X, Y) = -\,\text{Var}(X) = \boxed{-np(1-p)}.$$

The sign is negative because $X + Y = n$ is fixed: more successes means fewer failures.

(c) Let $n = 100$ and $p = 0.3$. Use a continuity-corrected Normal approximation to find $\mathbb{P}(Y - X > 30)$. (Hint: write the event in terms of X.) You may use $\Phi(0.98) \approx 0.8365$.

Solution. $Y - X = 100 - 2X > 30 \iff X < 35 \iff X \le 34$, since X is an integer. Here $\mathbb{E}[X] = 30$ and $\text{SD}(X) = \sqrt{100(0.3)(0.7)} = \sqrt{21} \approx 4.58$. The continuous boundary between 34 and 35 is 34.5:

$$z = \frac{34.5 - 30}{4.58} \approx 0.98, \qquad \mathbb{P}(Y - X > 30) \approx \Phi(0.98) \approx \boxed{0.837}.$$

(d) Repeat part (c) without a continuity correction, using 35 as the boundary (you may use $\Phi(1.09) \approx 0.8621$). The exact binomial probability is 0.8371. Compare the two approximations.

Solution. Without the correction, $z = \frac{35-30}{4.58} \approx 1.09$ and the approximation is $\Phi(1.09) \approx 0.862$, an error of about 0.025. With the correction the error is about 0.0006. The correction places the boundary in the gap between the integers 34 and 35 rather than on one of them.

6. Let $\overline{X}_n$ be the sample mean of an independent random sample of size n from a population with mean μ and variance σ^2.

(a) Compare the sampling distribution of $\overline{X}_n$ with the population distribution when $n = 1$ and when $n = 1000$. Discuss shape, center, and spread.

Solution. For $n = 1$, $\overline{X}_1 = X_1$, so the sampling distribution is the population distribution: same shape, center μ, spread σ.

For $n = 1000$ the center is still μ (exact, by linearity) and the spread shrinks to $\sigma/\sqrt{1000} \approx 0.032\sigma$ (exact, by independence). The shape is approximately Normal by the CLT, whatever the shape of the population, provided the variance is finite.

(b) A consumer agency weighs a simple random sample of 89 flour packages. The sample mean weight is 0.902 kg and the sample standard deviation is 0.122 kg. Calculate the estimated standard error of the sample mean. How does it differ from 0.122?

Solution. The estimated standard error replaces the unknown σ with s:

$$\widehat{\text{SE}}(\overline{X}) = \frac{s}{\sqrt{n}} = \frac{0.122}{\sqrt{89}} \approx \boxed{0.0129}\ \text{kg}.$$

The value 0.122 describes the spread of individual packages. The estimated SE describes the sample-to-sample variability of the mean. Neither one is the realized error of this sample mean.

(c) An employee plots a histogram of the 89 measurements, finds that it is not close to any Normal curve, and says this casts doubt on the Normal approximation for the sample mean. Comment.

Solution. The CLT concerns the standardized *sample mean*, not the individual observations, and it does not require the data to be Normal. A non-Normal histogram is therefore expected and is not evidence against the approximation by itself. The relevant question is whether $n = 89$ is large enough for this population's skewness and tails, which a histogram can inform but does not settle on its own.

7. In a population, a fraction p of individuals have $x = 0$ and a fraction $1-p$ have $x = 2$, where $0 \le p \le 1$. Let $X_1, \dots, X_n$ be a random sample with replacement from this population.

   (a) What is the exact sampling distribution of $\overline{X}_2 = (X_1 + X_2)/2$? (The answer depends on p.)

   Solution. The two draws are independent, so

$\overline{x}_2$	0	1	2
$\mathbb{P}(\overline{X}_2 = \overline{x}_2)$	p^2	$2p(1-p)$	$(1-p)^2$

   (b) What is the approximate sampling distribution of $\overline{X}_n$ for $n = 100$?

   Solution. One draw has $\mathbb{E}[X] = 2(1-p)$ and $\mathrm{Var}(X) = 4(1-p) - 4(1-p)^2 = 4p(1-p)$. By the CLT,

   $$\overline{X}_{100} \dot{\sim} \mathsf{N}\Big(2(1-p),\ \tfrac{4p(1-p)}{100}\Big),$$

   that is, standard deviation $2\sqrt{p(1-p)}/10$.

   (c) Is $\overline{X}_n/2$ an unbiased estimator of $1-p$ for all $n \ge 1$? Explain.

   Solution. Yes. By linearity,

   $$\mathbb{E}\Big[\tfrac{\overline{X}_n}{2}\Big] = \tfrac{1}{2n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \tfrac{1}{2n}\, n \cdot 2(1-p) = 1-p$$

   for every $n \ge 1$. Unbiasedness is a statement about the center of the sampling distribution and holds exactly at every sample size.

8. Let f be a function on $[0,1]$ with $0 \le f(x) \le 1$, and let θ be the area under f over $[0,1]$. We want to estimate θ. Let (X, Y) be a pair of independent Uniform$(0,1)$ random variables, so (X, Y) is a randomly chosen point in the unit square, and the probability of a region equals its area. Define

   $$W = \begin{cases} 1 & \text{if } Y \le f(X), \\ 0 & \text{otherwise.} \end{cases}$$

   Let (X_i, Y_i, W_i), $i = 1, 2, \dots$ be independent copies of (X, Y, W) and let $\overline{W}_n = \frac{1}{n}\sum_{i=1}^{n} W_i$.

   (a) What are $\mathbb{P}(W = 1)$ and $\mathbb{E}[W]$?

   Solution. $W = 1$ exactly when the point falls under the curve, so $\mathbb{P}(W = 1) = \text{area} = \theta$. Since W is an indicator, $\mathbb{E}[W] = \mathbb{P}(W = 1) = \theta$.

   (b) If n is large, is $\overline{W}_n$ a good estimator of θ? Explain.

   Solution. Yes. By the law of large numbers, $\overline{W}_n$ concentrates around $\mathbb{E}[W] = \theta$: for every fixed $\varepsilon > 0$, $\mathbb{P}(|\overline{W}_n - \theta| > \varepsilon) \to 0$. In addition, $\mathbb{E}[\overline{W}_n] = \theta$ for every n, so it is unbiased.

   (c) Construct an unbiased estimator of $3\theta + 7$. What is its variance when $n = 1000$? (The answer depends on θ.)

   Solution. Take $3\overline{W}_n + 7$. Then $\mathbb{E}[3\overline{W}_n + 7] = 3\theta + 7$. Because W_i is Bernoulli(θ), $\mathrm{Var}(\overline{W}_n) = \theta(1-\theta)/n$, so

   $$\mathrm{Var}(3\overline{W}_n + 7) = 9\,\frac{\theta(1-\theta)}{n} = \boxed{\frac{9\,\theta(1-\theta)}{1000}}.$$

   (d) Is $\cos(\overline{W}_n)$ an unbiased estimator of $\cos(\theta)$? Explain, and give a concrete case.

   Solution. No, in general. Expectation passes through linear functions only, and $\mathbb{E}[g(T)] \ne g(\mathbb{E}[T])$ for nonlinear g. For example, with $n = 1$ and $\theta = \frac{1}{2}$,

   $$\mathbb{E}[\cos W_1] = \tfrac{1}{2}\cos 0 + \tfrac{1}{2}\cos 1 \approx 0.770, \qquad \cos(\theta) = \cos(0.5) \approx 0.878.$$

9. In a population, the quantitative variable x takes the value 0 for 20% of individuals, 2 for 15%, and 4 for 65%. A random sample with replacement of size n is drawn.

   (a) For $n = 2$, find the sampling distribution of the sample maximum M.

   Solution. $M = 0$ only for $\{0,0\}$: probability $0.20^2 = 0.04$. $M = 2$ for $\{2,2\}$, $\{0,2\}$, $\{2,0\}$: $0.15^2 + 2(0.20)(0.15) = 0.0225 + 0.06 = 0.0825$. Otherwise $M = 4$: $1 - 0.04 - 0.0825 = 0.8775$. Thus

m	0	2	4
$\mathbb{P}(M=m)$	0.04	0.0825	0.8775

(b) Is the sample maximum an unbiased estimator of the population maximum? Explain.

Solution. No. For $n=2$, $\mathbb{E}[M] = 2(0.0825) + 4(0.8775) = 3.675 \neq 4$, a bias of -0.325. For any n there is positive probability, 0.35^n, of drawing no 4 at all; then $M < 4$, and $M \leq 4$ always, so $\mathbb{E}[M] < 4$. The sample maximum is biased downward at every n.

(c) How large must n be so that the sampling distribution of M puts probability more than 99% on the value 4?

Solution. $\mathbb{P}(M=4) = 1 - \mathbb{P}(\text{no } 4) = 1 - 0.35^n$, which exceeds 0.99 when $0.35^n < 0.01$. Since $0.35^4 \approx 0.0150$ and $0.35^5 \approx 0.0053$,

$$\boxed{n \geq 5}.$$

10. A population is described by two parameters θ_1 and θ_2. From a simple random sample of size n, an econometrician has an unbiased estimator T_1 of θ_1 and an unbiased estimator T_2 of θ_2, with variances σ_1^2 and σ_2^2.

(a) Is $T_1 - 3T_2$ necessarily an unbiased estimator of $\theta_1 - 3\theta_2$? Explain.

Solution. Yes. By linearity, $\mathbb{E}[T_1 - 3T_2] = \theta_1 - 3\theta_2$. Independence of T_1 and T_2 is not needed.

(b) Is $\text{Var}(T_1 - 3T_2) = \sigma_1^2 + 9\sigma_2^2$ necessarily true? Explain.

Solution. No. In general $\text{Var}(T_1 - 3T_2) = \sigma_1^2 + 9\sigma_2^2 - 6\,\text{Cov}(T_1, T_2)$, and the shortcut needs $\text{Cov}(T_1, T_2) = 0$ (for example, independence). Counterexample: let $T_2 = T_1/3$, which is unbiased for $\theta_2 = \theta_1/3$. Then $T_1 - 3T_2 = 0$ and its variance is 0, while $\sigma_1^2 + 9\sigma_2^2 = 2\sigma_1^2 > 0$.

(c) Is $T_1^2 + T_2$ necessarily an unbiased estimator of $\theta_1^2 + \theta_2$? Explain.

Solution. No. Squaring is not linear. Since $\mathbb{E}[T_1^2] = \text{Var}(T_1) + \theta_1^2$,

$$\mathbb{E}[T_1^2 + T_2] = \theta_1^2 + \sigma_1^2 + \theta_2,$$

so the bias is σ_1^2, which is positive unless T_1 has no variability.

(d) Now suppose that for large n, T_1 and T_2 are approximately Normal and independent. Is $T_1 - 3T_2$ approximately Normal? What is its approximate distribution?

Solution. Yes. A linear combination of independent Normal variables is Normal:

$$T_1 - 3T_2 \;\dot{\sim}\; \mathsf{N}\big(\theta_1 - 3\theta_2,\ \sigma_1^2 + 9\sigma_2^2\big).$$

Independence is used here for both the joint Normality and the variance formula, which connects back to part (b).

11. In a population, a fraction p of individuals have $x = 1$ and the remaining fraction $1 - p$ have $x = 3$. Let $X_1, \ldots, X_n$ be the values of x for n individuals sampled with replacement.

(a) Is $3 - \dfrac{1}{2n}\displaystyle\sum_{i=1}^{n} X_i$ an unbiased estimator of p? Explain.

Solution. One draw has $\mathbb{E}[X_i] = p + 3(1-p) = 3 - 2p$, so

$$\mathbb{E}\Big[3 - \tfrac{1}{2n}\textstyle\sum X_i\Big] = 3 - \tfrac{3-2p}{2} = \tfrac{3}{2} + p \neq p.$$

It is *not* unbiased; its bias is $\frac{3}{2}$. (The estimator $(3 - \overline{X})/2$ would be unbiased.)

(b) Is $\dfrac{9}{8} - \dfrac{1}{8n}\displaystyle\sum_{i=1}^{n} X_i^2$ an unbiased estimator of p? Explain.

Solution. One draw has $\mathbb{E}[X_i^2] = p + 9(1-p) = 9 - 8p$, so

$$\mathbb{E}\Big[\tfrac{9}{8} - \tfrac{1}{8n}\textstyle\sum X_i^2\Big] = \tfrac{9}{8} - \tfrac{9-8p}{8} = p.$$

Yes, it is unbiased.

(c) Suppose a single observation X is Normal with mean 1 and standard deviation σ. Construct an unbiased estimator of $7 + 4\sigma^2$ and show that it is unbiased.

Solution. Because the mean is known to be 1, $\mathbb{E}[(X-1)^2] = \text{Var}(X) = \sigma^2$. Therefore $7 + 4(X-1)^2$ is unbiased:

$$\mathbb{E}\big[7 + 4(X-1)^2\big] = 7 + 4\sigma^2.$$

12. A population consists of four individuals A, B, C, D with quantitative characteristic x equal to 0, 2, 4, 6, respectively. A simple random sample of size $n = 2$ is taken without replacement, with values X_1 and X_2.

    (a) What is the sampling distribution of $Y = X_1 + X_2$?

    Solution. There are six equally likely unordered samples:

Sample	AB	AC	AD	BC	BD	CD
Y	2	4	6	6	8	10

    so

y	2	4	6	8	10
$\mathbb{P}(Y = y)$	1/6	1/6	1/3	1/6	1/6

    (b) Is $Y/2$ an unbiased estimator of the population mean? Explain.

    Solution. $\mathbb{E}[Y/2] = \frac{1}{6}(1) + \frac{1}{6}(2) + \frac{1}{3}(3) + \frac{1}{6}(4) + \frac{1}{6}(5) = 3$, and the population mean is $\frac{1}{4}(0 + 2 + 4 + 6) = 3$. Yes, it is unbiased.

    (c) Calculate $\mathrm{Var}(Y/2)$ and compare it with $\sigma^2/2$, where σ^2 is the population variance of one uniformly chosen value. What explains the difference?

    Solution. The values of $Y/2$ are $1, 2, 3, 3, 4, 5$, with deviations from 3 of $-2, -1, 0, 0, 1, 2$, so

    $$\mathrm{Var}(Y/2) = \frac{4 + 1 + 0 + 0 + 1 + 4}{6} = \frac{5}{3}.$$

    The population variance is $\sigma^2 = \frac{9+1+1+9}{4} = 5$, so $\sigma^2/2 = 2.5$. The sample mean is less variable than the with-replacement formula suggests because sampling without replacement makes the two draws negatively dependent. The finite-population correction $\frac{N-n}{N-1} = \frac{2}{3}$ reproduces it: $2.5 \cdot \frac{2}{3} = \frac{5}{3}$.

13. Let X and Y be independent Normal random variables with $\mathbb{E}[X] = \mathbb{E}[Y] = \mu$, $\mathrm{SD}(X) = 2$, and $\mathrm{SD}(Y) = 5$. Consider two estimators of μ:

$$\widehat{\mu}_1 = \tfrac{1}{2}(X + Y), \qquad \widehat{\mu}_2 = 0.6X + 0.4Y.$$

    (a) Which estimator is preferable? Explain.

    Solution. Both are unbiased, since the weights sum to one. Their variances are

    $$\mathrm{Var}(\widehat{\mu}_1) = \tfrac{1}{4}(4 + 25) = 7.25, \qquad \mathrm{Var}(\widehat{\mu}_2) = 0.36(4) + 0.16(25) = 5.44.$$

    With zero bias, MSE = Var, so $\widehat{\mu}_2$ is preferable: it puts more weight on the more precise measurement.

    (b) Does your comparison in part (a) depend on X and Y being Normal? Explain.

    Solution. No. Unbiasedness uses only linearity of expectation, and the variance formulas use only independence and the two variances. Normality would matter only for probability statements about the estimators, such as tail probabilities.